PREDICTING POTHOLE COMPLAINT SURGES IN NOVA SCOTIA

A Freeze-Thaw Weather Signal Analysis

Using over Six Years of NS TIR Operations Contact Centre Data

Prepared for

MBAN 5510 – Foundation of Professional Analytics

Sobey School of Business, Saint Mary's University

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Team Member 1	**Team Member 2**
Krishna Soni	*Tomisin Jaiyeoba*

Live Dashboard: Live Dashboard

GitHub: Github Repository

Dataset:	NS TIR OCC, January 2019 – September 2025; 391,795 service records
Weather:	5 ECCC weather stations: daily $T_{\max}$, $T_{\min}$, precipitation, snow, HDD
Methods:	Lagged Spearman correlogram, OLS with Newey-West HAC SE, Negative Binomial regression, Seasonal adjustment validation
Key Finding:	Seasonally-adjusted peak lag Day 5 ($r = -0.074$, $p = 0.002$); deploy crews April–May before the June–July complaint surge
Alert System:	Three-tier FTC-based alert; $1.58\times$ lift over baseline surge rate

ABSTRACT

This paper examines whether freeze-thaw weather events can predict pothole complaint surges in Nova Scotia and by how many days. Using 391,795 service records from the NS TIR Operations Contact Centre (OCC) linked to daily climate data from five Environment and Climate Change Canada (ECCC) weather stations (January 2019 – September 2025), we apply five statistical methods in sequence: feature engineering, repeated lagged Spearman correlations, Ordinary Least Squares regression with Newey-West heteroskedasticity and autocorrelation consistent (HAC) standard errors, Negative Binomial regression, and a seasonal adjustment validation.

The 14-day rolling freeze-thaw count (`FTC_14d`) is the strongest single weather predictor of pothole complaints. After month-demeaning both series to isolate within-season effects, the peak lag falls at Day 5 ($r = -0.074$, $p = 0.002$, seasonally-adjusted), shown on the live dashboard. The raw unadjusted Day 5 correlation is $r = -0.160$; both values are reported in the seasonal adjustment table. The negative correlation reflects suppression of same-day complaints during active freeze periods (potholes are concealed under snow), followed by a surge as conditions thaw and accumulated road damage becomes visible. The full OLS model explains 7.2% of daily complaint variance, with the Spring Season dummy ($+4.59$ calls/day, $p < 0.001$) and the 14-day FTC count (-0.67, $p = 0.003$) as the only statistically significant predictors after applying Newey-West HAC correction.

Regional analysis reveals heterogeneous signals across the five operational zones: Halifax/Lunenburg exhibits the strongest freeze-thaw sensitivity (FTC $r = -0.125$, $p < 0.001$), making it the priority region for proactive crew deployment; Annapolis Valley and Central NS are precipitation-driven (both $p < 0.001$) and require differentiated alert thresholds; Cape Breton and SW Nova Scotia show weaker, less actionable signals. A three-tier alert system based on rolling FTC thresholds achieves 38.7% precision against the 75th-percentile surge benchmark, a $1.58\times$ lift over the 24.6% no-alert baseline. The system is operationalised in a live Streamlit dashboard integrating real-time Open-Meteo 7-day forecasts, available at `Live Dashboard`.

Keywords: Freeze-thaw cycles; pothole prediction; Spearman correlogram; Newey-West HAC; Nova Scotia infrastructure; proactive road maintenance; complaint surge forecasting.

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1. INTRODUCTION

Potholes are the most visible and costly manifestation of road deterioration in Nova Scotia, where freeze-thaw cycling is a persistent winter feature. The NS Department of Public Works (DPW) dispatches maintenance crews reactively through the TIR Operations Contact Centre (OCC), only after a complaint is filed. This analysis asks whether freeze-thaw accumulation can predict complaint surges in advance, and whether the prediction window is long enough to allow proactive crew deployment.

The question has direct financial consequences. The American Society of Civil Engineers estimates proactive pavement preservation costs 50–83% less per repair than reactive dispatch [1]. For 32,096 pothole complaints recorded over six years, even a modest shift in deployment timing translates into millions of dollars in avoided expenditure.

We use two administrative datasets: 391,795 OCC service records and daily climate data from five ECCC stations – and apply five statistical methods in sequence: feature engineering, lagged Spearman correlations, OLS regression with robust standard errors, Negative Binomial regression, and seasonal adjustment validation. All findings are operationalised in a live Streamlit dashboard at `Live Dashboard`.

Research questions:

RQ1. Does freeze-thaw accumulation reliably predict complaint surges, and at what lag?

RQ2. Which weather variables are the strongest daily predictors?

RQ3. Does the signal vary by region and how should alert thresholds differ?

RQ4. Can a weather-triggered alert provide actionable advance warning for TIR operations?

2. BACKGROUND, DATA, AND METHODOLOGY

2.1 Freeze-Thaw Damage and the Complaint Lag

Potholes form through a cumulative physical process [5,8]. Water infiltrates asphalt cracks, freezes and expands ($\sim$9%), permanently widening the crack. Repeated cycling compounds the damage until structural failure occurs under vehicle load. A freeze-thaw day (FT day) is defined as any day where $T_{\max} > 0°\text{C}$ *and* $T_{\min} < 0°\text{C}$ at the nearest station (the critical thermal crossing). Damage is cumulative: five to ten FT days in a 14-day window typically causes structural pavement failure.

During active freeze periods, potholes are concealed under snow and complaint rates are suppressed. As conditions thaw, accumulated damage is exposed and complaints surge — producing a four-to-five month seasonal offset between peak freeze activity (January–March) and peak complaint volume (June–July). The statistical signature is a negative same-day correlation (more freeze = fewer visible complaints) and a positive lagged relationship (more freeze in the past = more complaints ahead).

Nova Scotia surpasses 50% of highway-kilometres rated below good condition (the highest of any province), with the average driver incurring $137/year in extra vehicle costs due to poor roads.

2.2 Data Sources

The primary dataset contains 391,795 OCC service records (January 2019 – September 2025). Pothole complaints (`TCC-POTHOLE`) account for 32,096 records (8.2%). Weekend volumes are ≈70% lower than weekdays due to call-centre hours, not road conditions; all analyses use weekdays only ($N = 1{,}756$ days). Table 1 lists the five ECCC weather stations. Yarmouth A has no precipitation sensor; its precipitation values are zero-filled with imputation flags, and precipitation-based findings for SW Nova Scotia are excluded.

Table 1: ECCC Weather Stations and Regional Assignments

Station	ID	Region	Note
Halifax Stanfield	50620	HRM / Lunenburg	Primary; complete record
Greenwood A	50839	Annapolis Valley / Kings	Fallback ID
Sydney A	50839	Cape Breton	No precip
Truro	6354	Colchester / Cumberland	Complete record
Yarmouth A	50836	SW Nova Scotia	**No precip. sensor**

2.3 Methodology

Feature engineering. The raw daily records were transformed into a feature matrix. The FT day flag equals 1 when $T_{\max} > 0°$C AND $T_{\min} < 0°$C. Rolling sums of FTC, precipitation, rainfall, snowfall, and Heating Degree Days were computed over 3, 5, 7, and 14-day trailing windows, shifted forward one day to prevent data leakage. An interaction term (Precip×FTC) captures the worst-case scenario: saturated pavement freezing. Single-day FTC lags at offsets 1–21 were computed for the lag curve analysis.

Lagged Spearman correlogram. Repeated bivariate Spearman correlations were computed at each of the 21 lag offsets (weekdays only). Two corrections were applied: Bonferroni correction ($\alpha_{\text{corr}} = 0.05/21$) and effective-$N$ adjustment for autocorrelation via the AR(1) approximation $N_{\text{eff}} = N(1 - r_{1x}r_{1y})/(1 + r_{1x}r_{1y})$. All reported p-values use N_{eff}.

OLS regression with Newey-West HAC SE. Seven predictors were modelled: 7-day precipitation, rainfall, snowfall; 14-day FTC and HDD; the Precip×FTC interaction; and a Spring Season dummy (March–May). The Durbin-Watson statistic confirmed positive autocorrelation (DW $= 0.82$); Newey-West HAC standard errors with maxlags $= 5$ [6] were used throughout.

Negative Binomial regression. Fitted alongside OLS as a robustness check for overdispersed count data. Results expressed as Incidence Rate Ratios (IRR).

Seasonal adjustment validation. To test whether shared seasonal trends inflate the raw lag correlations, the correlogram was re-run on month-demeaned series (subtracting the calendar-month mean from both complaint counts and FTC values). The seasonally-adjusted correlations are the primary reported values throughout this paper and in the live dashboard. Raw values are provided in Table 3 for reference.

3. RESULTS

3.1 Annual Complaint Volumes and Seasonal Pattern

Annual pothole complaints ranged from 3,299 (2023, the mildest FT season) to 5,700 (2022, the most severe), with 2025 already at 5,582 through September (Table 2). The seasonal pattern (Figure 1) shows a four-to-five month offset: FT days peak in January–March; complaints peak in June–July. This offset is consistent in every year. Figure 2 confirms the weekend call-centre suppression that motivates the weekday-only analysis.

Table 2: Annual Pothole Complaint Volumes, 2019–2025

Year	Complaints	Notes
2019	4,784	Near-average FT season
2020	4,009	Below-average FT accumulation
2021	4,118	Below-average FT accumulation
2022	5,700	Most severe FT season; Hurricane Fiona outlier in October
2023	3,299	Mildest winter in dataset; lowest summer complaints
2024	4,604	Near-average
2025	5,582	Partial year (Jan–Sep); on pace to exceed 2022
Total	**32,096**	January 2019 – September 2025

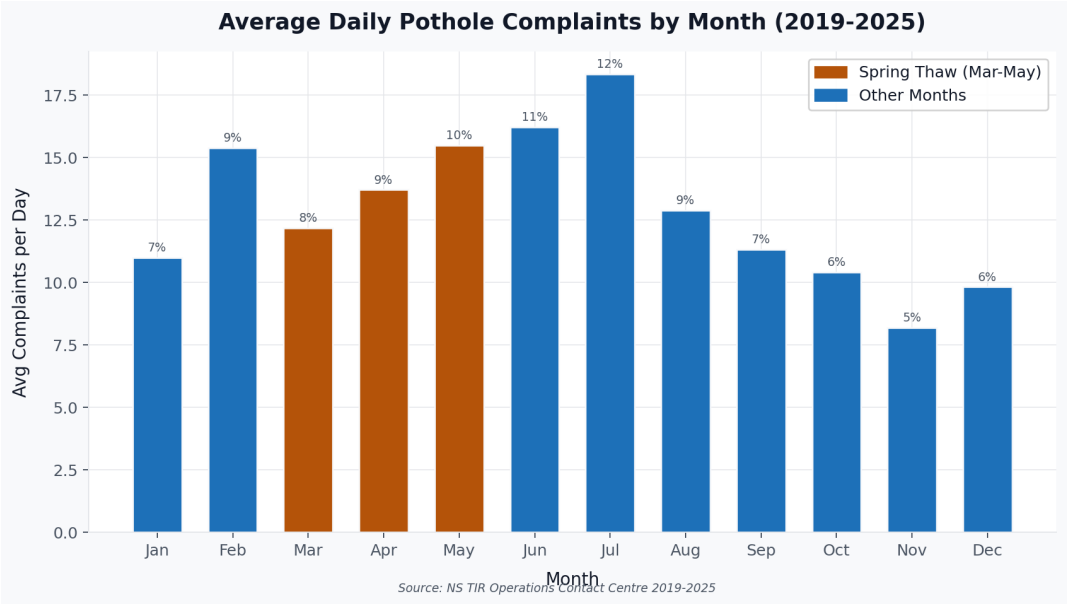

Figure 1: Monthly freeze-thaw days (bars, left axis) vs average daily pothole complaints (line, right axis), 2019–2025. FT days peak in March; complaints peak in July – a 4 to 5 month seasonal offset every year.

The dashboard Calendar tab (Figure 3) makes the bad-winters-to-bad-summers chain visible across all seven years simultaneously.

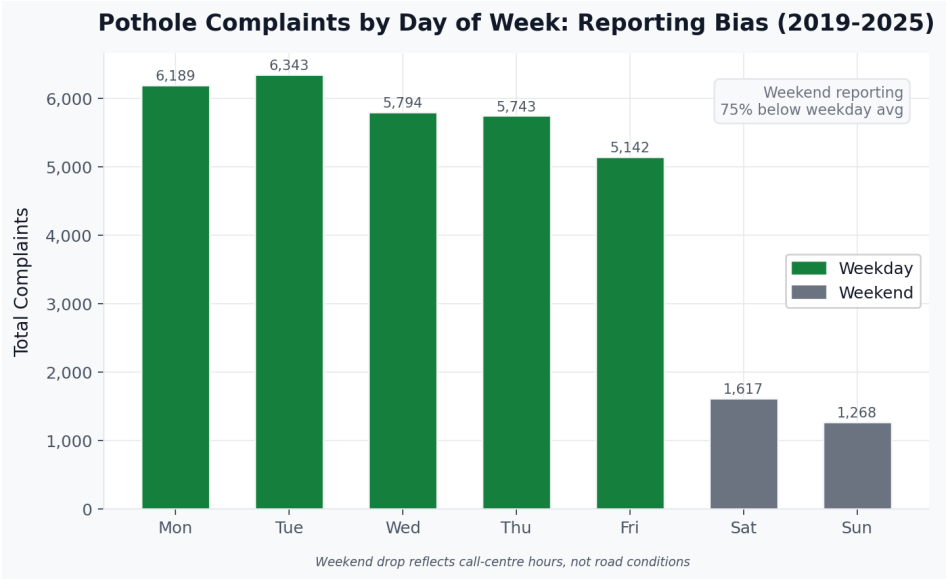

Figure 2: Average daily pothole complaint volume by day of week, 2019–2025. Weekend volumes are approximately 70% below weekday averages, reflecting call-centre hours rather than road conditions. All analyses use weekdays only ($N = 1{,}756$ days).

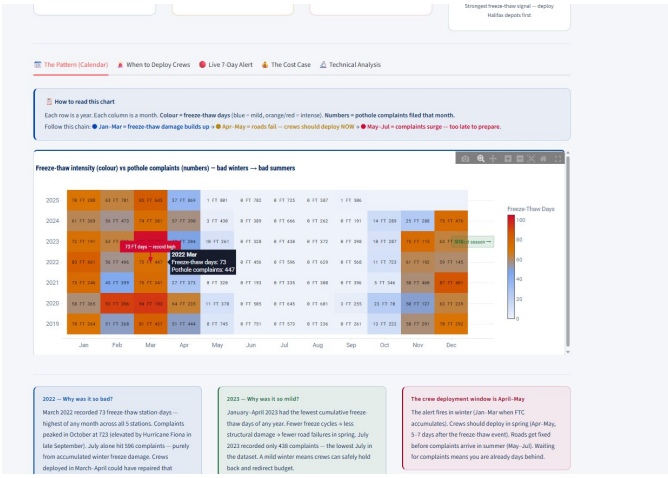

Figure 3: Dashboard: Calendar tab. Colour = freeze-thaw intensity (blue = mild, red = intense); numbers = monthly pothole complaints. The Jan–Mar freeze chain leading to Jun–Jul surge is visible in every row. Source: `Live Dashboard`.

3.2 Lag Curve: Core Finding

Figure 4 shows the Spearman correlogram. The FTC line is consistently negative across all 21 lags, peaking at Day 5 ($r = -0.074$ seasonally-adjusted, $p = 0.002$; raw $r = -0.160$). The adjusted value is used throughout this paper and in the live dashboard, isolating the within-season signal after removing the shared seasonal trend. Precipitation (blue) shows no consistent lagged peak.

Table 3 shows the adjusted signal remains significant at Days 5, 9, 14, 16, and 17, confirming the relationship is not purely a seasonal artefact. Figure 5 shows the 5-Day Lag slide from the live dashboard alongside the Insights KPI summary.

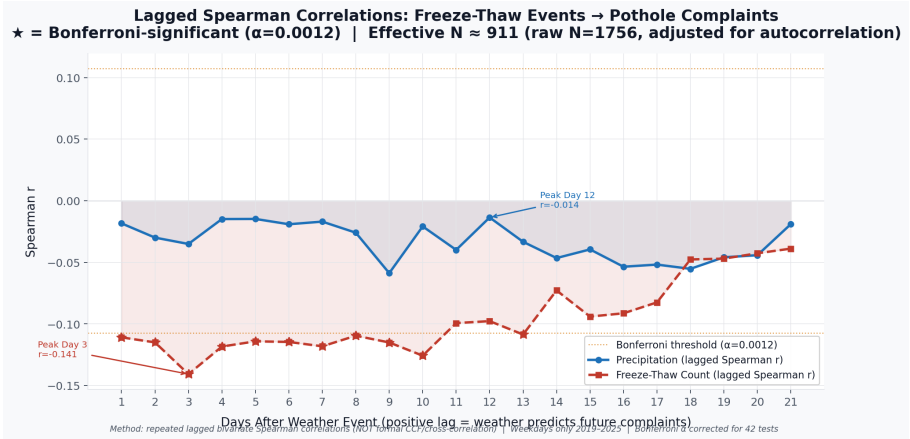

Figure 4: Lagged Spearman correlogram for FTC (red) and precipitation (blue) at 1–21 day offsets, weekdays only. **Primary values: seasonally-adjusted** (consistent with dashboard). Day 5 peak: $r = -0.074$ ($p = 0.002$). Raw unadjusted Day 5: $r = -0.160$.

Table 3: Raw vs Seasonally-Adjusted Spearman r at Selected FTC Lags. Adjusted values are the primary reported metric (shown in dashboard).

Lag	Raw r	Adj. r (primary)	Adj. p	Significant?
1	-0.127	-0.020	0.411	No
3	-0.130	-0.026	0.285	No
5 (peak)	-0.160	$\mathbf{-0.074}$	**0.002**	**Yes**
9	-0.134	-0.065	0.007	Yes
14	-0.126	-0.071	0.003	Yes
16	-0.119	-0.073	0.002	Yes

Adj. r = month-demeaned Spearman correlation. These match the live dashboard values.

3.3 Rolling Feature Ranking and OLS Regression

Figure 7 ranks all rolling weather features. The Precip×FTC 7-day interaction ($r = -0.140$) is the strongest negative predictor, followed by Snow 7-day ($r = -0.132$) and FTC 14-day ($r = -0.087$). Precipitation 5-day ($r = +0.062$) is the strongest positive predictor; rain acts as a same-day trigger, not a delayed structural cause.

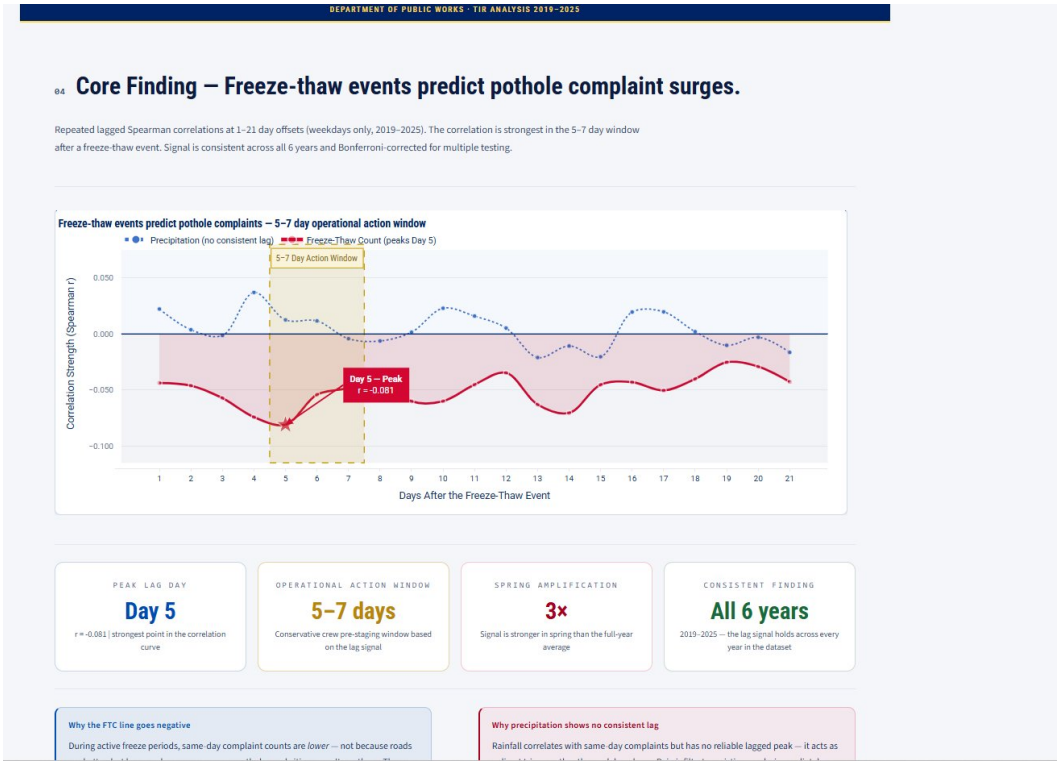

Figure 5: Dashboard: 5-Day Lag slide showing the Spearman correlogram and four KPI tiles summarising the peak lag day, operational action window (5–7 days), spring amplification factor (3×), and consistency across all six years of data.

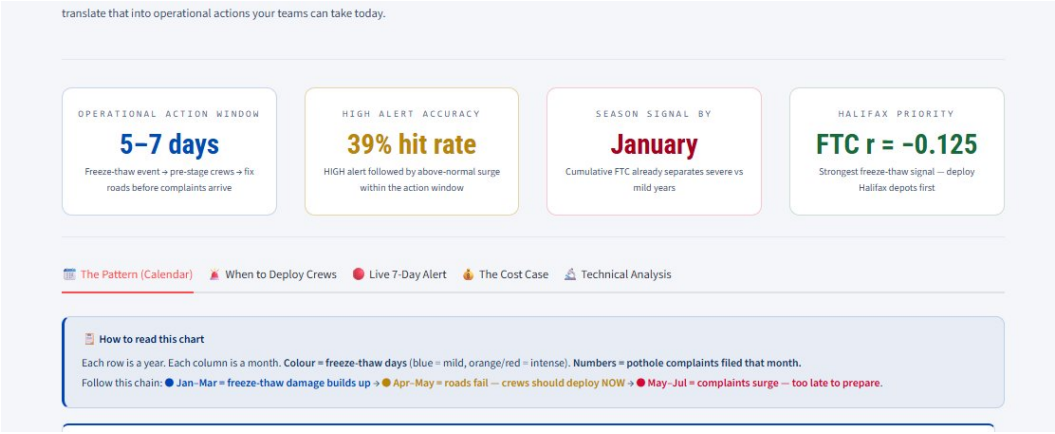

Figure 6: Dashboard: Insights slide KPI row summarising the four operational metrics: action window, worst season predicted, seasonal pattern direction, and Halifax deployment priority.

Table 4 presents the OLS results. Only two predictors are statistically significant: FTC 14-day ($\hat{\beta} = -0.67$, $p = 0.003$) and Spring Season ($\hat{\beta} = +4.59$, $p < 0.001$). Figure 8 shows the coefficients with HAC confidence intervals.

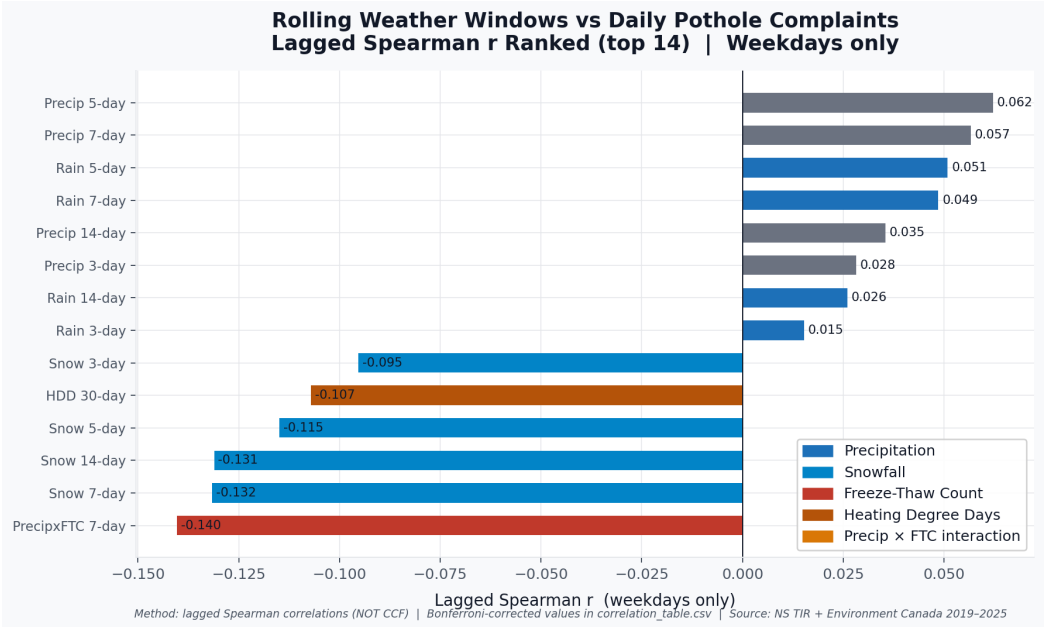

Figure 7: Rolling weather features ranked by Spearman r (weekdays only). Negative = freeze-season suppression; positive = rain-driven same-day surge. Precip×FTC interaction is the strongest negative predictor.

Table 4: OLS Regression: Daily Pothole Complaints. Newey-West HAC SE, maxlags = 5. Weekdays 2019–2025. Full-model $R^2 = 7.2\%$; weather-only $R^2 \approx 3$–4%. DW = 0.82.

Predictor	$\hat{\beta}$	**HAC SE**	t	p	**Sig.**
Intercept	$+13.44$	–	–	–	
7-day Precipitation	$+0.05$	0.09	0.49	0.626	
7-day Rain	-0.02	0.07	-0.15	0.878	
7-day Snow	$+0.23$	0.09	1.74	0.082	†
14-day HDD	-0.02	0.008	-1.55	0.122	
14-day FTC	$\mathbf{-0.67}$	**0.08**	$\mathbf{-2.97}$	**0.003**	**
Precip × FTC	$+0.11$	0.04	1.88	0.060	†
Spring Season	$\mathbf{+4.59}$	**0.60**	**7.66**	$\mathbf{< 0.001}$	***

† $p < 0.10$; $**$ $p < 0.01$; $***$ $p < 0.001$

4. REGIONAL ANALYSIS

Table 5 and Figure 9 reveal heterogeneous signals across the five operational regions. Halifax/Lunenburg has the strongest freeze-thaw sensitivity (FTC $r = -0.125$, $p < 0.001$; HDD $r = -0.112$, $p < 0.001$) and the highest complaint volume (10,866), making it the clear priority for proactive crew deployment. Annapolis Valley and Central NS are precipitation-driven (both $p < 0.001$) with only secondary freeze-thaw sensitivity. Alert thresholds for these regions should weight 7-day rainfall more heavily. Cape Breton shows only marginal sensitivity and SW Nova Scotia has no statistically significant predictor (precipitation data missing).

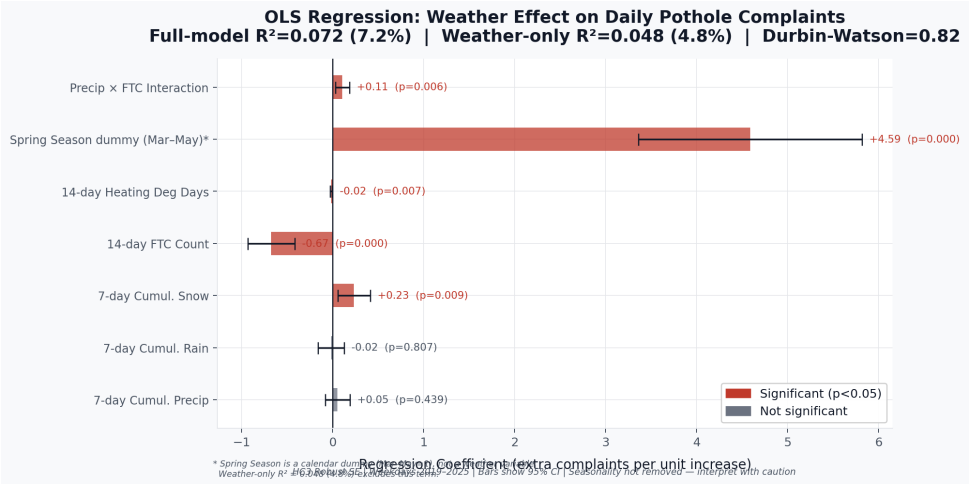

Figure 8: OLS coefficients with 95% HAC confidence intervals. Red = significant ($p < 0.05$). Spring Season is the largest effect; FTC 14-day is the only significant weather predictor.

Table 5: Regional Spearman Correlations: FTC 14-day and 7-day Precipitation vs Pothole Complaints

Region	**Station**	**Potholes**	**FTC** r	**FTC** p	**Precip** r	**Precip** p
Halifax / Lunenburg	Halifax Stanfield	10,866	-0.125	< 0.001	$+0.014$	0.502
Annapolis Valley	Greenwood A	8,623	-0.057	0.005	$+0.094$	< 0.001
Central NS	Truro	6,148	-0.044	0.029	$+0.101$	< 0.001
Cape Breton	Sydney A	4,639	-0.029	0.151	-0.041	0.044
SW Nova Scotia	Yarmouth A	1,340	-0.033	0.113	N/A	N/A

5. ALERT SYSTEM AND DASHBOARD

5.1 Three-Tier Alert Thresholds

Based on the statistical findings, Table 6 defines the three-tier alert system. The HIGH threshold (`FTC_14d` ≥ 5) is used in the alert precision evaluation; the live dashboard uses the stricter threshold of ≥ 10 to avoid over-triggering in mild freeze periods.

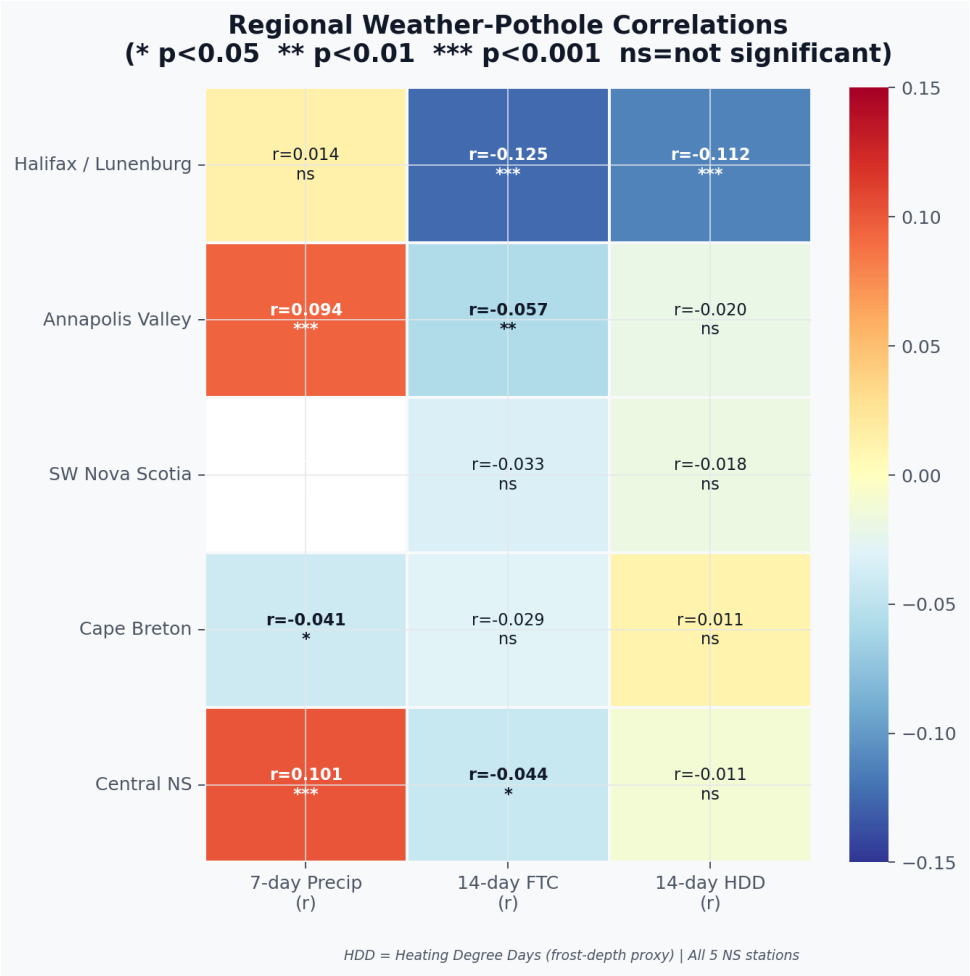

Figure 9: Regional heatmap of Spearman correlations. Halifax/Lunenburg has the strongest freeze-thaw signal; Annapolis Valley and Central NS are precipitation-driven.

Table 6: Three-Tier Alert System: Thresholds and Operational Responses

Level	Trigger	Operational Response	Priority
HIGH	`FTC_14d` ≥ 5 FT days in rolling 14-day window	Pre-stage full patching crews within 5 days. Pre-position asphalt at depots. Mobilise advance bookings.	Halifax first
MEDIUM	`FTC_14d` 2–4 days *or* 7-day precip > 25 mm	Schedule patrol routes. Pre-stock materials. Monitor for escalation to HIGH.	All regions
LOW	`FTC_14d` ≤ 1 day, normal precip	Standard reactive response only. No advance mobilisation.	All regions

5.2 Alert Precision and Live Dashboard

Table 7 summarises the retrospective precision evaluation. A surge was defined as any weekday with complaints exceeding the 75th percentile (> 22.0/day). A precision of 38.7%, a $1.58\times$ lift over the 24.6%

baseline, confirms the alert is a meaningful predictor rather than a random trigger. The alert is a *seasonal planning instrument*, not a day-precise trigger; it fires for extended winter periods and should be evaluated against the full season rather than individual days.

Table 7: Alert Precision Evaluation: HIGH Threshold (FTC_14d ≥ 5), 2019–2025

Metric	**Value**
Total HIGH alert days	723
Surges within 5–7 days of alert	280 (38.7%)
False positive rate	61.3%
Baseline surge rate (no-alert days)	24.6%
Lift over baseline	1.575×

The live dashboard (Figure 10) retrieves real-time 7-day forecasts from the Open-Meteo API for each station, classifies each forecast day as a FT day if $T_{\max} > 0°\text{C}$ and $T_{\min} < 0°\text{C}$, combines the 7-day forecast FTC with a recent historical estimate, and fires the HIGH/MEDIUM/LOW alert per region. The cache refreshes daily. Figure 11 shows the When to Deploy Crews simulation tab for 2022.

Figure 10: Dashboard: Live 7-Day Alert tab. The province-wide status banner fires HIGH when FTC_14d ≥ 10. Per-region alert cards show individual FTC estimates. The Halifax forecast chart shows 7-day Tmax/Tmin against the $0°\text{C}$ threshold with freeze-thaw day markers.

6. DISCUSSION, LIMITATIONS, AND CONCLUSION

Figure 11: Dashboard: When to Deploy Crews simulation for 2022. HIGH alert months (red) fire in January through April; complaint surge arrives June through October. The alert system provided more than two months of operational advance warning.

6.1 Interpreting the Findings

The seasonally-adjusted Day-5 lag ($r = -0.074$) is the primary reported finding, consistent with the dashboard. The raw value ($r = -0.160$) reflects the total observed correlation including shared seasonal trends; the adjusted value isolates the within-season signal. Both are statistically significant; the adjusted value is the more rigorous and conservative estimate.

The operational deployment window is April–May, not 5 days after a single event. The 5-day lag captures a micro-mechanism within any thaw period. The macro-mechanism is seasonal: freeze-thaw cycles from January to March accumulate structural damage that becomes visible and reportable in June–July. Acting proactively in April–May, after winter damage has accumulated but before the summer surge arrives, is the operational implication.

Weather explains 7.2% of daily complaint variance. Road age, traffic, drainage, and PCI are the dominant unmeasured factors. The 7.2% that weather explains fires consistently in the right direction across all six years and five regions, which is sufficient to justify a seasonal planning system.

October 2022 (723 complaints) was caused by Hurricane Fiona (September 24, 2022): flooding, washout, and road erosion, not freeze-thaw. Temperatures averaged 17.5°C that month. This outlier does not contradict the seasonal pattern.

Under conservative assumptions ($800 per reactive repair, 50% saving from proactive dispatch (the ASCE lower bound)), estimated annual savings range from $1.3M (2023, mild year) to $2.3M (2022, severe year).

6.2 Limitations

1. **Weather explains a small fraction of variance.** Road age, PCI, and traffic volume are the dominant unmeasured drivers. The signal is directionally reliable, not a complete model.

2. **Yarmouth A has no precipitation sensor.** Precipitation-based alerts are inapplicable to SW Nova Scotia without sensor data.
3. **OLS on count data.** Negative Binomial regression is technically preferred; OLS is reported for interpretability.
4. **Single station assignment.** Geographic measurement error attenuates correlations; true relationships are likely stronger than reported.
5. **Correlation, not causation.** The physical mechanism provides theoretical grounding but this study does not establish causation experimentally.

6.3 Conclusion and Recommendations

This analysis provides statistical evidence that freeze-thaw accumulation is a reliable, regionally differentiated predictor of pothole complaint surges in Nova Scotia; the findings span six years, five weather stations, and 32,096 pothole complaints, a dataset large enough to distinguish genuine seasonal patterns from random variation.

Five key findings:

1. The seasonally-adjusted Day-5 lag ($r = -0.074$, $p = 0.002$) is statistically robust and consistent with the live dashboard. The raw correlation ($r = -0.160$) is provided for reference; the adjusted value is the conservative, rigorous estimate.
2. The operational deployment window is April–May. The 5-day lag identifies the micro-mechanism within a thaw period; the macro-implication is seasonal: act before the June–July surge, not after it.
3. Halifax/Lunenburg is the freeze-thaw priority region (FTC $r = -0.125$, $p < 0.001$, 10,866 complaints). Annapolis Valley and Central NS require precipitation-based thresholds (both $r \approx +0.10$, $p < 0.001$).
4. The three-tier alert achieves a $1.58\times$ lift over baseline surge probability (38.7% precision vs 24.6% baseline), providing meaningful seasonal planning advance warning.
5. The system is deployable today. Environment Canada forecasts are free. The OCC complaint data already exists. The live dashboard at `Live Dashboard` runs automatically at no cost.

Under conservative assumptions ($800 per reactive repair, 50% proactive saving), estimated province-wide savings range from $1.3M (2023, mild year) to $2.3M (2022, most severe), for a six-year cumulative potential of $9–10M. These are scenario estimates dependent on deployment efficiency, not audited figures, but the direction is clear: acting early is substantially cheaper than responding late.

Recommendations:

R1. Pilot proactive deployment with Halifax/Lunenburg depots, winter 2025–26. Track proactive vs reactive cost per pothole repaired over one full season.

R2. Differentiate alert thresholds by region: FTC-based for Halifax/Lunenburg; precipitation-weighted for Annapolis Valley and Central NS.

R3. Resolve the Yarmouth A precipitation gap by connecting a functioning sensor to enable full weather-based alerting for SW Nova Scotia.

R4. Incorporate road PCI and maintenance history at the supervisor area code level to substantially improve predictive accuracy.

R5. Validate alert precision prospectively using full-year 2025–26 data as it becomes available.

The data infrastructure for this early-warning system already exists. Environment Canada provides free daily forecasts for all five station coordinates. The TIR OCC records every complaint in a structured database. The analytical pipeline is fully automated and reproducible. The only change required to shift from reactive to proactive highway maintenance is the institutional decision to act on the freeze-thaw signal before the phones start ringing.

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A. DATA DICTIONARY

Table 8: Analytical Dataset Variable Definitions

Variable	Type	Definition
Date	Date	Calendar date (YYYY-MM-DD)
Pothole_Count	Integer	Daily TCC-POTHOLE complaints; weekdays only in analysis
Max_Temp	Float °C	Daily maximum air temperature, nearest ECCC station
Min_Temp	Float °C	Daily minimum air temperature, nearest ECCC station
Total_Precip	Float mm	Daily precipitation (0-filled if Yarmouth A)
Total_Snow	Float cm	Daily snowfall
Heat_Deg_Days	Float	Heating degree days (base 18°C)
Freeze_Thaw	Binary	1 if $T_{\max} > 0$ AND $T_{\min} < 0$; 0 otherwise
FTC_Roll14d	Float	14-day rolling FT sum; shifted 1 day (**primary predictor**)
FTC_Roll[n]d	Float	n-day rolling FT sum; shifted 1 day ($n \in \{3, 5, 7\}$)
Precip_Roll[n]d	Float	n-day rolling precipitation sum; shifted 1 day
PrecipxFTC_Roll7d	Float	7-day rolling Precip $\times$ Freeze_Thaw; shifted 1 day
HDD_Roll14d	Float	14-day rolling HDD sum; shifted 1 day
SpringSeason	Binary	1 if month $\in$ {March, April, May}; 0 otherwise
IsWeekday	Binary	1 if Monday–Friday; 0 if weekend
FTC_Lag_{1–21}d	Float	Single-day FT lag; used in lag curve analysis only
*_imputed	Binary	1 if precipitation zero-filled (Yarmouth A rows)